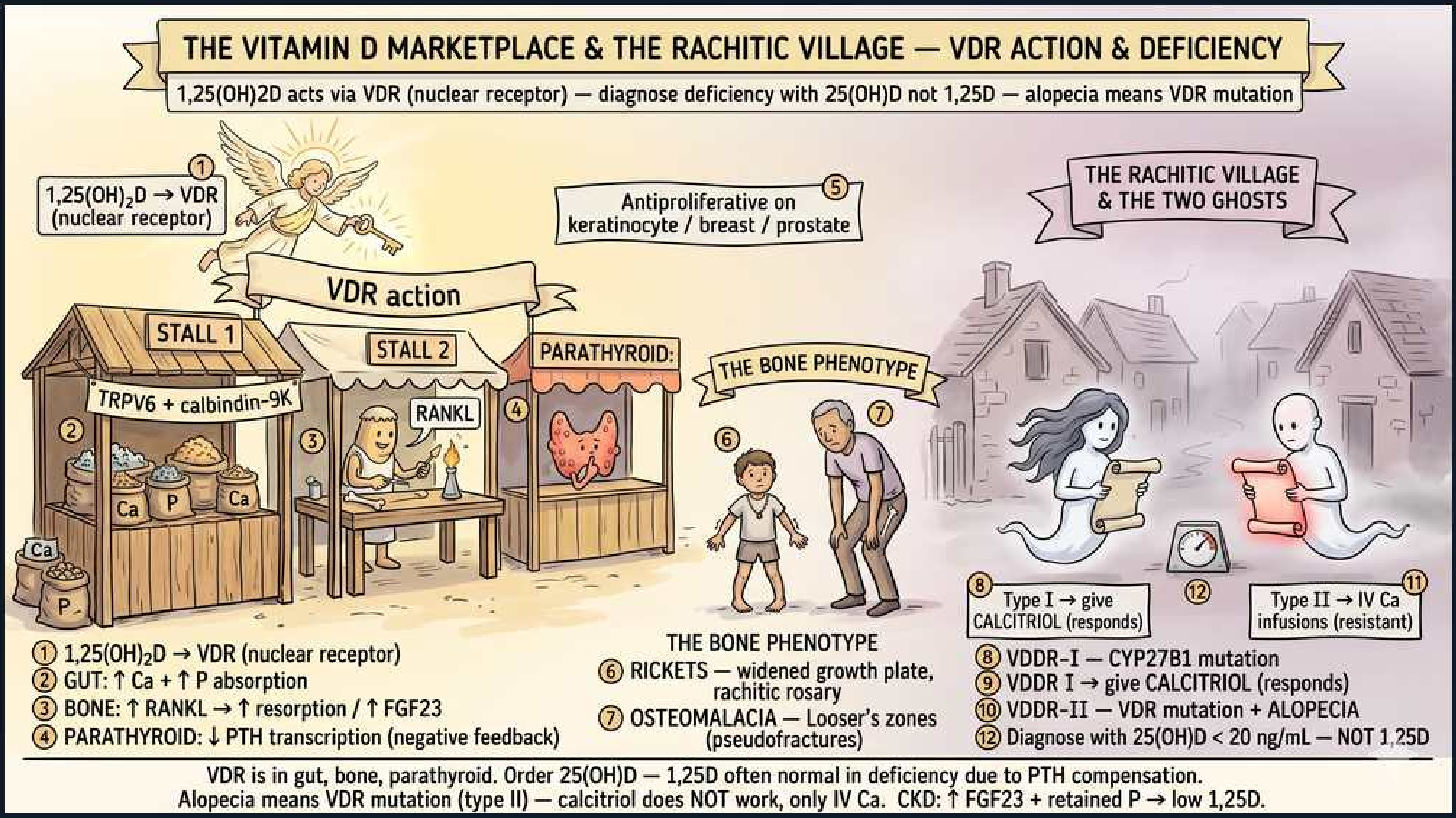
The Vitamin D Marketplace — VDR Action & Deficiency



1. 1,25(OH)2D acts via VDR (nuclear receptor)

WHAT YOU SEE

Banner: 1,25D acts via VDR nuclear receptor

WHAT IT ENCODES

Active 1,25-dihydroxyvitamin D (calcitriol) binds the intracellular vitamin D receptor, a nuclear transcription factor. Diagnose DEFICIENCY by measuring 25(OH)D (the storage form), NOT 1,25-D. VDR mutation causes end-organ resistance.

BOARD TRAP

Wrong answer: order 1,25-D to diagnose deficiency. Discriminator: 25(OH)D is the marker of vitamin D status; 1,25-D can be normal/high in deficiency due to PTH drive.

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2. 1,25(OH)2D → VDR signaling

WHAT YOU SEE

Angel holding 1,25D to VDR

WHAT IT ENCODES

Calcitriol is the hormonally active metabolite. Synthesis: skin/diet → 25-hydroxylation in liver → 1-alpha-hydroxylation in kidney (PTH-stimulated, FGF23-suppressed). It is the ligand for VDR-mediated transcription.

BOARD TRAP

Wrong answer: 25(OH)D is the active hormone. Discriminator: 1,25(OH)2D is active; 25(OH)D is the circulating storage/measured form.

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3. Gut: increase Ca and PO4 absorption

WHAT YOU SEE

Stall 1: gut, TRPV6 + calbindin-9K

WHAT IT ENCODES

In the intestine VDR upregulates TRPV6 and calbindin-9k to increase absorption of BOTH calcium and phosphate. This is the principal calcemic action of vitamin D — gut absorption, not bone.

BOARD TRAP

Wrong answer: vitamin D mainly works by pulling Ca from bone. Discriminator: its primary action is increasing GUT Ca and phosphate absorption.

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4. Bone: RANKL up, resorption, FGF23

WHAT YOU SEE

Bone stall: RANKL, resorption, FGF23

WHAT IT ENCODES

At high levels 1,25-D increases RANKL → osteoclastic resorption and stimulates osteoblast FGF23. In deficiency, inadequate mineralization causes rickets (children) or osteomalacia (adults).

BOARD TRAP

Wrong answer: vitamin D always builds bone. Discriminator: pharmacologic/high 1,25-D can drive resorption via RANKL; deficiency impairs mineralization.

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5. Parathyroid: suppresses PTH transcription

WHAT YOU SEE

Parathyroid stall, negative feedback

WHAT IT ENCODES

1,25-D directly suppresses PTH gene transcription (negative feedback) and the parathyroid expresses VDR and CaSR. Vitamin D deficiency therefore drives secondary hyperparathyroidism.

BOARD TRAP

Wrong answer: vitamin D stimulates PTH. Discriminator: 1,25-D INHIBITS PTH transcription; deficiency causes secondary hyperPTH.

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6. Antiproliferative — keratinocyte/breast/prostate

WHAT YOU SEE

Stall: antiproliferative on tissues

WHAT IT ENCODES

VDR signaling is antiproliferative and pro-differentiation in keratinocytes (psoriasis treatment with calcipotriol), breast and prostate. Explains non-classical actions of vitamin D analogs.

BOARD TRAP

Wrong answer: vitamin D only regulates calcium. Discriminator: VDR also has antiproliferative roles (basis for topical calcipotriol in psoriasis).

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7. Rickets — widened growth plate, rachitic rosary

WHAT YOU SEE

Bone phenotype: rickets, rosary

WHAT IT ENCODES

In children, defective mineralization of growth plates causes rickets: widened/cupped metaphyses, bowed legs, rachitic rosary at costochondral junctions, and frontal bossing. Low Ca/phosphate with high PTH and high alkaline phosphatase.

BOARD TRAP

Wrong answer: scurvy (vitamin C). Discriminator: rickets = vitamin D/mineralization defect with rachitic rosary and widened growth plates, elevated ALP.

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8. Osteomalacia — Looser zones

WHAT YOU SEE

Bone phenotype: osteomalacia, pseudofractures

WHAT IT ENCODES

In adults, undermineralized osteoid causes osteomalacia with bone pain, proximal myopathy, and Looser zones (pseudofractures) on imaging. Labs: low/normal Ca, low phosphate, high PTH, high ALP.

BOARD TRAP

Wrong answer: osteoporosis. Discriminator: osteomalacia shows Looser pseudofractures and high ALP from mineralization defect; osteoporosis has normal labs.

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9. Type I rickets — CYP27B1, give calcitriol

WHAT YOU SEE

Type I ghost, VDDR-I responds to calcitriol

WHAT IT ENCODES

Vitamin D-dependent rickets type I: loss of 1-alpha-hydroxylase (CYP27B1) → cannot make 1,25-D. Low 1,25-D. RESPONDS to physiologic calcitriol (bypasses the enzyme block).

BOARD TRAP

Wrong answer: type II. Discriminator: VDDR-I is enzyme deficiency (low 1,25-D) that RESPONDS to calcitriol replacement.

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10. Type II rickets — VDR mutation, alopecia

WHAT YOU SEE

Type II ghost, VDR mutation, alopecia

WHAT IT ENCODES

Vitamin D-dependent rickets type II: VDR mutation → end-organ RESISTANCE. 1,25-D is HIGH (compensatory) but ineffective. Alopecia is the clue. Requires high-dose calcium (IV) infusions, often resistant to calcitriol.

BOARD TRAP

Wrong answer: VDDR-I. Discriminator: type II has VDR mutation, HIGH 1,25-D, and ALOPECIA; resistant to calcitriol — give IV calcium.

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11. Alopecia = VDR mutation marker

WHAT YOU SEE

Alopecia note

WHAT IT ENCODES

Alopecia is the bedside discriminator for VDR (type II) resistance — it does not occur in type I. Reflects the role of VDR in hair follicle cycling independent of its ligand.

BOARD TRAP

Wrong answer: alopecia indicates type I. Discriminator: alopecia specifically signals VDR mutation (type II resistance).

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12. CKD: high FGF23, low 1,25-D

WHAT YOU SEE

Bottom banner: CKD FGF23 retained, low 1,25D

WHAT IT ENCODES

In CKD, phosphate retention raises FGF23 early, which suppresses 1-alpha-hydroxylase → low 1,25-D → secondary hyperparathyroidism and CKD-MBD. Treat with active vitamin D analogs and phosphate binders.

BOARD TRAP

Wrong answer: give plain cholecalciferol to fix CKD bone disease. Discriminator: CKD lacks 1-alpha-hydroxylase activity — give ACTIVE vitamin D (calcitriol/paricalcitol).

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13. Diagnose deficiency: 25(OH)D <20

WHAT YOU SEE

Diagnose with 25(OH)D <20, not 1,25D

WHAT IT ENCODES

Vitamin D deficiency is defined by 25(OH)D below ~20 ng/mL. In simple nutritional deficiency, 1,25-D is often normal or high due to secondary PTH-driven 1-alpha-hydroxylation — so it cannot diagnose deficiency.

BOARD TRAP

Wrong answer: low 1,25-D confirms deficiency. Discriminator: use 25(OH)D <20; 1,25-D may be normal/elevated in deficiency.
